Nikos Malakasis

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RESEARCH EXPERIENCE

Computation in Neural Circuits Lab @ Technical University of Munich (TUM) **Oct. 2022 – Present**
Doctoral Candidate *Munich, Germany*

- The Computation in Neural Circuits Lab is an interdisciplinary computational neuroscience lab led by Prof. Dr. Julijana Gjorgjieva.
- As a PhD candidate, I actively perform research and present my results at highly reputable conferences. I supervise thesis projects, teach classes, revise papers, and assist with grant writing.
    - My main research project focuses on the contribution of inhibition to the subcellular organization of synapses on dendrites through excitatory and inhibitory synaptic plasticity.
    - I have supervised four master's students and one bachelor's student on research projects.

Dendrites Lab @ Foundation for Research and Technology Hellas (FORTH) **Jul. 2020 – Sep. 2022**
Master's student Sep. 2020 – Dec. 2021 | Research Assistant, Dec. 2021 – Sep. 2022 *Crete, Greece*

- The Dendrites Lab is a computational neuroscience lab, led by Dr Yiota Poirazi. The lab's research focuses on elucidating dendrites' role in learning and memory.
- Using a bio-realistic Spiking Neural Network I implemented in C++, I studied how structural plasticity affects image classification. (See **Publications**)

Scientific Intern | Jul. 2020 – Aug. 2020

- During my summer internship in the lab, I worked on designing simplified interneuron models with active dendrites using Brian2.

Molecular Genetics of Aging Lab @ Academy of Athens(BRFAA) **Jul 2018 – Jul 2019**
Bachelor's student *Athens, Greece*

- The Molecular Genetics of Aging Lab is a molecular genetics lab led by Dr. Popi Syntichaki. The lab uses *C. elegans* as a model organism to study aging.
- I worked on identifying how the p38 MAPK pathway interacts with the mRNA decapping complex in *C. elegans*.

EDUCATION

Technical University of Munich **Ongoing**
Ph.D., Computational Neuroscience *Munich, Germany*

University of Crete **Dec 2021**
MSc, Bioinformatics *Crete, Greece*

- Graduated with 9.4/10 (excellent)

National and Kapodistrian University of Athens **Sep 2019**
BSc, Biology *Athens, Greece*

- Graduated with 7.25/10 (very good)
- Got an Erasmus+ scholarship to study for 5 months(*Feb - Jul 2018)* at **Universidade do Minho** in *Braga, Portugal*.

PUBLICATIONS AND NOTABLE CONFERENCE PRESENTATIONS

- **Publications:**
    - **N. Malakasis**, S. Chavlis and P. Poirazi, "Synaptic turnover promotes efficient learning in bio-realistic spiking neural networks", *2023 57th Asilomar Conference on Signals, Systems, and Computers*, Pacific Grove, CA, USA, 2023, pp. 942-949, doi: 10.1109/IEEECONF59524.2023.10476829.
    - I. Fritz, F. Wang, R. Chirif, **N. Malakasis**, J. Gjorgjieva, A. Ferreira Castro, "Synaptic density and relative connectivity conservation maintain circuit stability across development", *eLife* 14:RP108643; doi:

https://doi.org/10.7554/eLife.108643.1

- R. Rajan, R. F Dias, **N. Malakasis**, M. Baeta, X. Zhang, J. Gjorgjieva, L. Petreanu, "Visual experience exerts an instructive role on cortical feedback inputs to the primary visual cortex", Current Biology, 2026; 36, 1033-1044.e; doi: 10.1016/j.cub.2026.01.031

- **Notable Conference Presentations:**
  - Neuromatch 4.0; Flash Talk | AREADNE 2022; Poster Presentation: **Biologically Constrained Spiking Neural Network for Image Classification.**
  - Dendrites 2022; Poster Presentation: **Key Biological Features that help a Spiking Neural Network perform Image Discrimination.**
  - Bernstein 2023 | Cosyne 2024; Poster Presentation: **The Role of Inhibition in Shaping Dendritic Synaptic Arrangement.**
  - Dendrites 2024; Poster Presentation and Contributed Flash Talk: **The Role of Inhibition in Shaping Dendritic Synaptic Arrangement.**
  - Inhibition in the CNS Gordon Research Seminar and Conference 2025; Poster Presentation: **The Functional and Structural Role of Inhibition in Dendritic Synaptic Arrangement.**
  - Dendrites 2026; Poster Presentation: **The Structural Role of Inhibition in Shaping Synaptic Arrangement.**
  - Dendrites 2026; Poster Presentation: **A mechanistic model of sensory and feedback signal integration via dendritic computation.**

EXTRA COURSES, SKILLS & INTERESTS

- **Courses:**
  - Neuromatch Academy Deep Learning Summer Course 2022.
    - ➢ During this course, I worked with a team on a reinforcement learning project to mimic human performance on a behavioural task with reinforcement learning agents by training on human fMRI data.
  - OIST Computational Neuroscience Course 2024.
    - ➢ During this course, I worked alone on a self-proposed project to understand the mechanistic connectivity of the visual thalamocortical microcircuit that facilitates surprise.
- **Skills:** Adaptability, strategic planning, fast learning, supervising, programming in multiple languages, computational modeling, data analysis, data visualization, machine learning, deep learning, molecular biology techniques, bacterial cultures and cloning, confocal microscopy, *C.elegans* handling.
- **Interests**: Video games, generative AI art, graphics design, video editing, martial arts, snowboarding, strategic board games, trading card games, anime.